

Building a Generic Data Ingestion Pipeline

20 August 2026 07:02

What is Generic Pipeline

Generic Pipeline in ADF Or MS Fabric is a reusable, parametrized PL WHICH CAN PROCESS DATA FROM MULTIPLE SOURCE SYSTEM TO SINK SIDE WITHOUT CREATING A SEPARATE PL FOR EACH SOURCE

EX :-
SOURCE 1 : AZURE SQL DB
SOURCE 2 :- LAKEHOUSE
SOURCE 3 :- ADLS GEN2
SOURCE 4 :- SFTP SERVER

PRE-REQUISITES :- FOR AZURE SQL DB

SOURCE - AZURE SQL DB - MULTIPLE TABLE
USING :- DATA PIPELINE
SINK :- LAKEHOUSE

- 1. SOURCE
- 2. SINK
- 3. CONFIG FILE - WH
- 4. PL
- 5. LKP-FOREACH
 - a. COPY

Id	Src_schema	Src_table	Sink_folder	Sink_filename	Is_active
1	Saleslt	Customer	Sqlldb/bronze	Customer.csv	1
2	Saleslt	Address	Sqlldb/bronze	Address.csv	1

Username - intellibi
Pass - test@123

SFTP PASS- /ZHsYNa3lDyC0iEmgPM0CehSc7b7YxHx

CONNECTING STRING - adlssftp2008.user1@adlssftp2008.blob.core.windows.net
PORT - 22

SOURCE 1- AZURE SQL DB

Id	Src_schema	Src_table	Sink_folder	Sink_filename	Is_active
1	Saleslt	Customer	Sqlldb/bronze	Customer.csv	1
2	Saleslt	Address	Sqlldb/bronze	Address.csv	1

SOURCE 2 :- AZURE DATA LAKE STORAGE GEN2

ID	SRC_CONTAINER	SRC_FOLDER	SRC_FILENAME	SINK_FOLDER	SINK_FILENAME	IS_ACTIVE
1	GLOBAL	CUSTOMER	CUSTOMER.CSV	ADLS/BRONZE	CUSTOMER.CSV	1
2	GLOBAL	EMPLOYEE	EMPLOYEES.CSV	ADLS/BRONZE	EMPLOYEES.CSV	1

SOURCE 3 :- SFTP SERVER

ID	SRC_CONTAINER	SRC_FOLDER	SRC_FILENAME	SINK_FOLDER	SINK_FILENAME	IS_ACTIVE
1	GLOBAL-SFTP	INPUT	SALES.CSV	SFTP/BRONZE	SALES.CSV	1

MAPPING_TABLE

MAPPING_ID	PIPELINE_ID	SRC_SCHEMA	SRC_TABLE	SRC_CONTAINER	SRC_FOLDER	SRC_FILENAME	SINK_FOLDER	SINK_FILENAME	IS_ACTIVE	LOAD_TYPE	UPDATED_DATE
101	1	SALESLT	CUSTOMER	NULL	NULL	NULL	SQLDB/BRONZE	CUSTOMER.CSV	1	FULL	2026-08-20 07:00:00
102	1	SALESLT	ADDRESS	NULL	NULL	NULL	SQLDB/BRONZE	ADDRESS.CSV	1	FULL	2026-08-20 07:00:00
103	2	NULL	NULL	GLOBAL	CUSTOMER	CUSTOMER.CSV	ADLS/BRONZE	CUSTOMER.CSV	1	FULL	2026-08-20 07:00:00

104	2	NULL	NULL	GLOBAL	EMPLOYEE	EMPLOYEES.CSV	ADLS/BRONZE	EMPLOYEES.CSV	1	FULL	2026-08-20 07:00:00
105	3	NULL	NULL	GLOBAL-SFTP	INPUT	SALES.CSV	SFTP/BRONZE	SALES.CSV	1	FULL	2026-08-20 07:00:00

```
create table mapping_details(
mapping_id int,pipeline_id int,
src_schema varchar(50), src_table varchar(50), src_container varchar(50), src_folder varchar(50),
src_filename varchar(50), sink_folder varchar(50), sink_filename varchar(50), is_active int, load_type
varchar(50)
);
```

-- SOURCE -1 AZURE SQL DB

Insert into mapping_details values

```
(101,1,'saleslt','customer',null,null,null,'sqldb/bronze','customer.csv',1,'full_load'),
```

```
(102,1,'saleslt','address',null,null,null,'sqldb/bronze','address.csv',1,'full_load'),
```

-- SOURCE -2 ADLS GEN2

```
(103,2,null,null,'global','customer','customer.csv','adls/bronze','customer.csv',1,'full'),
```

```
(104,2,null,null,'global','employee','employees.csv','adls/bronze','employees.csv',1,'full'),
```

-- SOURCE -3 SFTP SERVER

```
(105,3,null,null,'global-sftp','input','sales.csv','sftp/bronze','sales.csv',1,'full')
```

```
;
```

SRC_DETAILS_TABLE

PIPELINE_ID	SRC_NAME	SRC_TYPE	DESCRIPTION	IS_ACTIVE	UPDATED_AT
1	SQLDB	DATABASE	FINANCE_DATA	1	2026-08-20 07:00:00
2	ADLS	FILE_SYSTEM	INSURANCE_DATA	1	2026-08-20 07:00:00
3	SFTP	FILE_SYSTEM	SALES_DATA	1	2026-08-20 07:00:00

```
create table src_details(
pipeline_id int,
src_name varchar(50),
src_type varchar(50),
description varchar(100),
is_active int
);
```

Insert into src_details values

```
(1,'sqldb','database','finance_data'1),
```

```
(2,'adls','file_system','insurance_data'1),
```

```
(3,'sftp','file_system','sales_data'1);
```

CREATE A JOIN OPERATION FOR ABOVE 2 TABLES

TABLE 1 - SRC_DETAILS S

TABLE 2 - MAPPING_DETAILS M

SELECT

s.pipeline_id,

s.src_name,

s.src_type,

s.description

m.src_schema,

m.src_table,

m.src_container,

m.src_folder,

m.src_filename,

m.sink_folder,

m.sink_filename,

m.load_type

From src_details s

Full outer join mapping_details m

On s.pipeline_id = m.pipeline_id

Where

s.is_active =1

And

m.is_active =1

And

s.pipeline_id= @pipeline_id

